

Aditya Raj

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EDUCATION

Indian Institute of Information Technology, Tiruchirappalli

Expected May 2027

Bachelor of Technology in Electronics and Communication Engineering (CGPA: 7.12 / 10) Tiruchirappalli, Tamil Nadu

ACHIEVEMENTS

- **LeetCode: Guardian** (Max Rating: **2192**, Top 0.2% Globally) – Ranked **155th** & **331st** in Weekly Contests 438 & 445 among 32,000+ participants (@adityaaarajjj).
- **Codeforces: Expert** (Max Rating: **1845**) – Ranked **316th** & **403rd** in Div. 2 (Rounds 1108 & 1064), and **133rd** in Div. 3 (@adityaraj18).
- **CodeChef 4★** (Max Rating: **1837**, @adityaaa_rajjj) and **6-Kyu at AtCoder** (Rating: **980**, @de_barr.).

TECHNICAL SKILLS

Languages: C++, Python, JavaScript (ES6+), TypeScript, Java, HTML5, CSS3, SQL

Frontend: React.js, Next.js, Redux, Tailwind CSS, Modern CSS, HTML Canvas, Vite

Backend & Databases: Node.js, Express.js, REST APIs, WebSockets, MongoDB, PostgreSQL

Tools & Concepts: Git, GitHub, Linux, Docker, VS Code, Postman, OOP, System Design, DBMS, OS, Computer Networks

PROJECTS

AI Task Manager | Node.js, Express.js, MongoDB, React, REST APIs, NLP

GitHub 

Full-Stack Intelligent Autonomous Task & Project Management Application

- Engineered an AI-powered project management platform featuring autonomous goal decomposition into structured milestones and subtasks.
- Implemented an NLP-driven Command Palette (Cmd+K) parsing natural language strings for instant task creation with priority, tags, and due dates.
- Built interactive workflows including HTML5 drag-and-drop Kanban board, 4-quadrant Eisenhower Matrix, and productivity velocity analytics.
- Integrated a Pomodoro focus engine with Web Audio API sound synthesis and created a modular Express REST backend with atomic data backups.

CodeAudit | React, Node.js, Express, Gemini, Docker, Redis, MongoDB

GitHub 

Real-Time AI-Powered Code Review & Repository Analysis Platform

- Built a full-stack platform for real-time GitHub repository analysis, combining AI-powered code review with execution-based code health analysis.
- Integrated Google Gemini for contextual code reviews, CWE detection, and repository-aware assistance with real-time WebSocket streaming.
- Engineered an asynchronous BullMQ + Redis pipeline with isolated Docker sandboxes for secure code execution with CPU, memory, timeout, and network restrictions.
- Implemented GitHub REST API integration with Redis ETag caching, MongoDB-backed analysis history, and historical Code Health Score tracking across commits.

Algorithm Visualizer | JavaScript (ES6+), HTML5 Canvas, CSS3, Data Structures

GitHub 

Interactive Algorithm Simulation & Data Structures Platform

- Developed interactive, step-by-step visualizations for complex sorting algorithms (Quick, Merge, Heap) and graph traversals (Dijkstra, BFS/DFS), breaking down abstract logic into intuitive visual state transitions.
- Engineered custom HTML5 Canvas animations with dynamic array/graph generators and step-by-step execution to help demystify tough algorithmic topics such as recursion trees, partitioning, and shortest-path relaxation.
- Integrated real-time comparative analytics displaying comparison counters, array access metrics, and theoretical time/space complexities for enhanced conceptual learning.

RELEVANT COURSEWORK

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|--------------------------------|---------------------|------------------------------|-----------------------|
| ● Data Structures & Algorithms | ● Operating Systems | ● Database Management (DBMS) | ● Computer Networks |
| ● Object-Oriented Programming | ● Machine Learning | ● Artificial Intelligence | ● Digital Electronics |